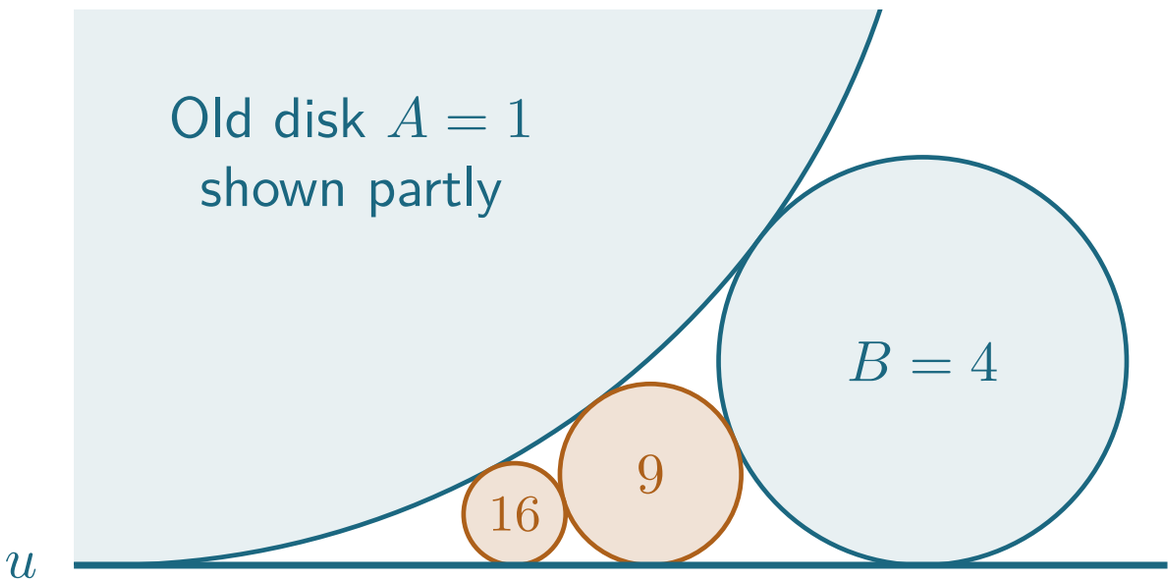
Independent depth at every initial bridge gap

Each insertion stays inside one original empty triangular gap.
All original disks and the marked cap stay fixed.



New curvatures in this example: $C_1 = 9$, $C_2 = 16$.

$$\sqrt{C_n} = \sqrt{B} + n\sqrt{A}$$

An exact translation in square-root curvature, for every chain length.

$$(2j-1) \longmapsto (2j-1, 2j)^{m_j}(2j-1)$$

Apply separately before each retained first-layer box.

$R = 3k - 3 + 2\sum_{j=1}^{k-1} m_j$ independent twist regions.

All widths and depth vectors are fixed before the quantum level grows.